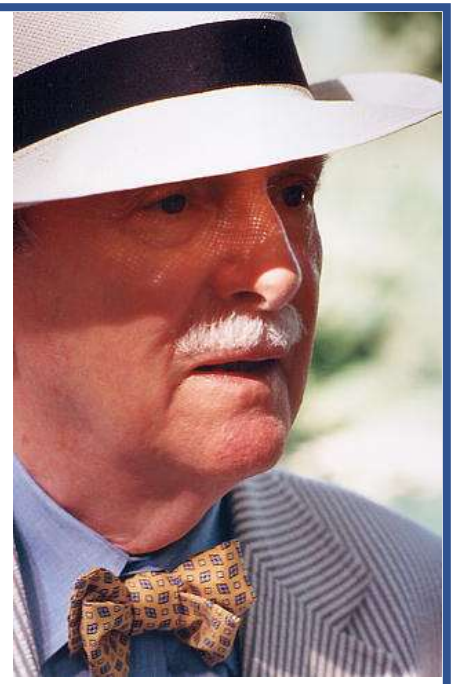
GFN 1000
In Dialogue with Nature

Nathan Sivin

Texts and Core Question



Text 10:

10a: Nathan Sivin: Why the Scientific Revolution Did Not Take Place in China – or Didn't It? (An academic paper)

10b: Shen Kua: *Brush Talks from Dream Brook* 沈括《梦溪笔谈》

Core Question:

How should we compare Western science and Chinese science?

- How do we understand science in non-Western traditions?
- What is the "Scientific Revolution" ?

Today…

- Introduction
  - Authors
  - Needham's question
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  V. Fallacies of Historical Reasoning (3 fallacies; Para 35-46)

Nathan Sivin (席文)

- Sinologist, historian, and writer.
- Born in 1931 in America.
- 1958, S.B. in humanities with a chemistry minor, MIT.
- 1960, 1966, M.A. and Ph.D. in the history of science, Harvard U.
- Used to be the Professor Emeritus of Chinese Culture and of the History of Science at the University of Pennsylvania.



1931-2022

According to his website…

Language Ability

Chinese, speaking and reading: excellent.
Classical Chinese, reading: excellent.
Japanese, speaking and reading: good.
French, reading: good.
German, reading: fair.

Courses Taught

Chinese Literature
Chinese Science and Natural Philosophy
Classical Chinese
Classics of Science and Medicine
Current Sinology
Health Care in Traditional and Contemporary China
Introduction to the Far East
Introduction to the History of Science
Medicine and the Body (research seminar)
Methods of Research in Sinology
Methods of Research in the History of Chinese Medicine
Readings in Chinese Science
Science and Philosophy from Antiquity to Copernicus
Science, Medicine and Ritual
Scientific Revolution
Seminar in Chinese Culture
Seminar in Organizational Culture (anthropological field work course)
Seminar in the Scientific Revolution and Enlightenment
Professions (sociology field work course)
Taoism (history of Taoist religious movements)
Various experimental and collaborative courses

Nathan Sivin (席文)



1931-2022

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- 1960, 1966, M.A. and Ph.D. in the history of science, Harvard U.
- Used to be the Professor Emeritus of Chinese Culture and of the History of Science at the University of Pennsylvania.
- Coauthor of *Science and Civilization in China* Vol. 5 & 6.
- Major areas of study: the history of science and technology in China, Traditional Chinese Medicine, Chinese philosophy, and Chinese religious beliefs.

Selected publications:

- *Chinese Alchemy(炼金术) Preliminary Studies*, Harvard University Press, 1968
- *Cosmos and Computation in Early Chinese Mathematical Astronomy(天文学)*, Leiden: E. J. Brill, 1969
- *Science and Civilization in China*. Vol. 5, Part 4. Chemical Discovery.
- *Medicine, Philosophy and Religion in Ancient China*. Researches and Reflections. Variorum Collected Studies Series. 1995
- *The Way and the Word: Science and Medicine in Early China and Greece*, Yale University Press, 2002
- *Health Care in Eleventh-Century China*, Springer, 2015

“Why the Scientific Revolution Did Not Take Place in China — or Didn't It?” *Chinese Science* 5 (1982): 45–66.

-A response to Needham’s Question

Praise

"[These three papers are] the most stimulating contributions to the underlying problems of comparative history of science that I have ever seen."
(H. Floris Cohen, *The Scientific Revolution: A Historiographical Inquiry*, p. 472)

- The three papers include Text 10a and another paper by Sivin.

Why does Sivin use Shen Kua as his case?



沈括 (1031-1095)

- A polymath (博学者) “scientist” and politician in Song dynasty.
- One of the most versatile figures in the history of Chinese science and engineering.
- His achievements in geology, mathematics, meteorology, zoology, pharmacology, *etc.* positioned him among the greatest scientific minds of the time.
- His research interests were connected to responsibilities and commitments of his **civil service appointments**.

In the 11th century, literati (文人学士) like Shen Kua started to take interest in all sorts of practical affairs. Especially, technical matters that earlier were fit only for clerks (文书) or artisans (工匠).

Civil Service Life of Shen Kuo



---The connection between Shen's life as an active statesman (政治家) and his wide-ranging inquiries.

- 1050s, in his early 20s, Shen Kuo received several minor local posts: designing and superintending (监管) a drainage (排水) and embankment (路堤) system.
- 1063, Shen Kuo passed the national examinations and received a degree (进士). He became a professional financial administrator.
- 1072, he was appointed as Supervisor of the Directorate of Astronomy (司天监) and undertook calendar reform.
- 1073, Shen Kuo involved in the New Policies (新政) of Wang Anshi (王安石), including water control, military preparations, and local administrations.
- 1075, Shen became the head of the Finance Commission (三司使).

Brush Talk from Dream Brook or 梦溪笔谈

- Taxonomy (分类学) of knowledge.
- 609 jottings (笔记) and 17 categories.
- Understanding the ‘classification of knowledge’ of Shen Kuo.
- One of the most valuable resources in the history of science in China.



故事	Ancient Usages
辩证	Philological Criticism
乐律	Music and Mathematical Harmonics
象数	Regularities Underlying the Phenomena
人事	Human Affairs
官政	Civil Service
权智	Wisdom in Emergencies
艺文	Literature
书画	Calligraphy and Painting
技艺	Technical and Artful Skills
器用	Noble or Ancient Artifacts
神奇	The Supernormal
异事	Strange Occurrences
谬误	Errors
讥谑	Wit and Satire
杂志	Miscellaneous
药议	Materia Medica

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The Needham Question

Why did modern science emerge in Western Europe rather than in China, *despite* the earlier predominance of Chinese science and technology?

- "Why did modern science, the mathematization of hypotheses about Nature, with all its implications for advanced technology, take its meteoric (急速的) rise only in the West at the time of Galileo?" "Why modern science had not developed in Chinese civilization . . . ?"

- "Why, between the first century B.C. and the fifteenth century A.D., Chinese civilization was much more efficient than occidental (西洋) in applying human natural knowledge to practical human needs."

Before Needham

- 任鸿隽 (1886-1961), “The Reason for China’s Lack of Science”(《说中国无科学之原因》) published in *Science* in 1915 (Footnote 8).

  “即使这样，欧洲有科学不过是近三四百年的事情，说我国古代没有科学，也没有什么问题。”



Hongjuan Ren
(1886-1961)

- 1944年，竺可桢,《中国古代为什么没有产生自然科学？》

我国历史上环境不适宜的缘故。在《科学与技术》期刊上，陈立先生《我国科学不发达之心理分析》一文里的结论是：“中国科学之不发达，我曾溯源于（一）拟人思想的泛生论；（二）没有工具思想的直观方法；（三）没有逻辑；（四）没有分工；（五）客观与主观的混淆；（六）理智的不诚实等等。但这一切我都指出，系反映着客观社会的组织。在宗法阶段的社会，便只有宗法社会的思想。”去



Kezhen Zhu
Chinese geologist, meteorologist.

Why did Modern Science not Emerge in China?

Our possible answers…

- Examination system
- Confucianism (儒教)
- Agricultural society
- ….

Why did Modern Science not Emerge in China?

Needham's answer

- "In sum, I believe that the analyzable differences in social and economic pattern between China and Western Europe will in the end illuminate… both the earlier predominance of Chinese science and technology and also the later rise of modern science in Europe alone." (SCC7:2 p. 23)

- Needham suggested it was the rise of the bourgeoisie (城市资产阶层) (rich people who owned property) in Western Europe from 15th century that gave rise to science.

Emergence of Modern Science in the West

Rise of Bourgeoisie (资产阶级)

- four collapses: feudal order, States, Church and Mongol Empire (封建秩序, 国家, 教会和蒙古帝国)
- the bourgeoisie had more freedom in doing trades
- development of mathematics for transactions (交易)

Information revolution

- moveable type printing ➡ books were printed in vernacular (白话) languages
- mathematization became a general practice
- common people could have education
- the society became knowledgeable

Not everyone agrees with Needham

Nathan Sivin
Serious criticism of Needham Question
7 shaky assumptions + 3 fallacies(谬论)!

Critical thinking

- Critical thinking is “Reasonable reflective thinking focused on deciding what to believe or do.” (Robert Ennis) It is used in problem solving and decision making.

- Critical thinking is the analysis of available facts, evidence, observations, and arguments to form a judgement.

Core critical thinking skills

- Cognitive skills
  - interpretation
  - analysis
  - evaluation
  - inference
  - explanation
  - self-regulation
  - …
- Mental dispositions
  - Inquisitiveness
  - Truth-seeking
  - Open-mindedness
  - Systematicity
  - Self-confidence
  - Cognitive maturity
  - …

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Two Conclusions (Par. 1)

'Why the Scientific Revolution Did Not Take Place in China
– or Didn't It?'

This essay makes the case for two conclusions. First, why the scientific revolution did not take place in China is not a question that historical research can answer. It becomes a useful question primarily when one locates the fallacies that lead people to ask it. Second, a scientific revolution, by the criteria that historians of science use, did take place in China in the eighteenth century. It did not, however, have the social consequences that we assume a scientific revolution will have. (Para 1)

Two Assumptions in Needham Question

Second question: why, between the first century B.C. and the fifteenth century A.D., Chinese civilization was much more efficient than occidental in applying human natural knowledge to practical human needs?

- The earlier predominance of Chinese science and technology:

  two assumptions:

  - Application of Chinese science in daily life
  - Superiority of Chinese science over western science

Two Assumptions Refuted (Para. 4-8)

- Assumption 1: Application of Chinese science in daily life
  - what was applied to human needs was not Chinese science but technology
- Assumption 2: Superiority of Chinese science over western science
  - out of question; a meaningful comparison involves many aspects such as science, politics and sociology

Are These Science according to Sivin?



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The absence of unity in Chinese science

“The natural knowledge that was being applied to human needs was not what we usually call Chinese science” (para. 4)

“Early technology did not succeed or fail according to how well it applied the insights of early science” (para. 5)

“One aspect was that there does not seem to have been a systematic connection between all the sciences in the minds of the people who did them” (para. 10)

“I would have to say that I failed to find the internal unity of Chinese science that I was looking for in the mind of Shen Kua” (para. 17)

Brush Talk from Dream Brook or 梦溪笔谈

- Taxonomy (分类学) of knowledge.
- 609 jottings (笔记) and 17 categories.
- Understanding the ‘classification of knowledge’ of Shen Kuo.



故事	Ancient Usages
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艺文	Literature
书画	Calligraphy and Painting
技艺	Technical and Artful Skills
器用	Noble or Ancient Artifacts
神奇	The Supernormal
异事	Strange Occurrences
谬误	Errors
讥谑	Wit and Satire
杂志	Miscellaneous
药议	Materia Medica

Examples of three categories

- Technical and Artful Skills 技艺：
  - Including mathematics, architecture, games and their winning strategies, and good skills of various kinds.
    - total number of possible situations in chess 棋局
    - moveable type printing 活版印刷
- Strange Occurrences 异事：
  - “strange objects” , “change” 变化
    - the rainbow 虹
    - Fossil 化石
    - Tornado 龙卷风
- Miscellaneous 杂志：
  - Things couldn’t be classified or categorized.
    - Transformation of the land and the sea 海路变迁
    - the compass 指南针
    - Petroleum 石油

Why the Scientific Revolution Did Not Take Place in China — or Didn't It?

Sivin's response to Needham's Question:

- The natural knowledge that was being applied to human needs was not what we usually call Chinese science (para. 4)
- Early technology did not succeed or fail according to how well it applied the insights of early science (para. 5)
    - Early science vs early technology
        - Early science: members of the small educated class in China who did science and passed down their understanding in books.
        - Early technology: a matter of craft (工艺) and manufacturing skills that artisans (工匠) privately transmitted to their children and apprentices (学徒).
- Sivin criticizes comparing the entirety of ancient science and technology in one civilization to another because it is too general (para. 6)

Why the Scientific Revolution Did Not Take Place in China — or Didn't It?

- There was not a "systematic connection between all the sciences in the minds of people who practiced them" (para. 10) because there was
  - No integration of a natural philosophy,
  - No development of scientific methodology,
  - No organization of scientific activity similar to what developed in Western Europe in science academies and universities, which were precursors to a scientific revolution.
    - 1603, The Lincean Academy, Rome
    - Francis Bacon, constructing a community of scientific workers. (*The New Atlantis*,1626)
    - 1660, the Academie des Sciences in Paris
    - 1662, the Royal Society in London

“They had sciences but no science, no single conception or word for the overarching sum of all of them.”

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Scientific Revolution

"I use 'Scientific Revolution' to refer primarily to the transition in the exact sciences between Galileo and Laplace and its wider repercussions by 1800."
(Footnote 1) (Various definitions)

- Galileo Galilei (1564–1642)
- Isaac Newton (1642–1727) (Text 3a, 3b)
- Pierre-Simon Laplace (1749–1827)

  a French scholar and polymath whose work was important to the development of engineering, mathematics, statistics, physics, astronomy, and philosophy.
- The birth of modern science (mathematization + experimentation)(c.f Text 3a, 3b)

Why something did NOT happen?

It is striking that this question—Why didn't the Chinese beat Europeans to the Scientific Revolution?—happens to be one of the few questions that people often ask publicly about why something didn't happen in history. It is analogous to the question of why your name did not appear on page 3 of today's newspaper. (Para. 18)



- no direct answer (or infinitely many answers)
- heuristic question 启发式的问题
- this heuristic question holds an urgency arising from the shaky assumptions

Why did modern science emerge in Western Europe rather than in China, despite the earlier predominance of Chinese science and technology?

Shaky assumptions behind this Question:

- A1: Everybody ought to have Science Revolution (Every culture: include China!)
- A2: Everybody wanted to have the Science Revolution (Modern science is a good thing: Chinese want!)
- A3:The Western model of the Scientific Revolution **is a standard pattern for all capable civilizations.** It will lead to similar institutional and social changes (Western Europe is the model! And other culture should have similar consequences)

Why did modern science emerge in Western Europe rather than in China, despite the earlier predominance of Chinese science and technology?

Faith/Conviction behind this Question:

- A4: European civilization has some intrinsic fitness (Superiority of European: intelligence, imagination, talent, virtue)

- A5: Science is universal, objective, and value-free

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Universal and value-free? Wishful thinking

32 Beyond the narrow, abstract realm in which exactitude is possible, values and subjective judgments come to bear on every activity situated within a society. There are, for instance, profound differences between the character of modern scientific activity in the contemporary People's Republic of China and United States. They reflect different predominant convictions about the relations between basic and applied science, the relation of both to general culture, the roles of scientists in defining research programs, procedures for planning and supporting individuals' research projects, expectations about the social aims to which scientific work will contribute, the organization and status of professional scientists, the connections of political ideas and scientific knowledge, and the division of national resources between science and other priorities, and between various scientific activities. That

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Historical Reasoning: Three Fallacies

- F1: an earlier state of culture is confused as a necessary condition (文化早期的情况与必要条件混为一谈)
- F2: the absence of the subsequent state is confused with an inhibiting factor (没有发生继之而来的应当发生的状况与抑制因素混为一谈)
- F3: some historians look at intellectual factors alone, while some at socio-economic factors alone

Fallacy 1: Example (Para. 36-37)

- Chinese astronomers "were not interested in applied technical sciences, e.g., in developing theoretical tools which could be used to control the flight of a cannon shell or to direct ships safely across the sea." → no unified scientific system
- "Interested in applied technical sciences" is a necessary condition for "developing unified scientific system"
- But the same lack of interest is prominent in the impetus theoreticians from John Philoponus (fl. ca. 530) to Jean Buridan (ca. 1295–ca. 1358) (See also Text 2) → Galilean mechanics

Fallacy 1: no interest in applied technical sciences → no modern science

Historical Reasoning: Three Fallacies

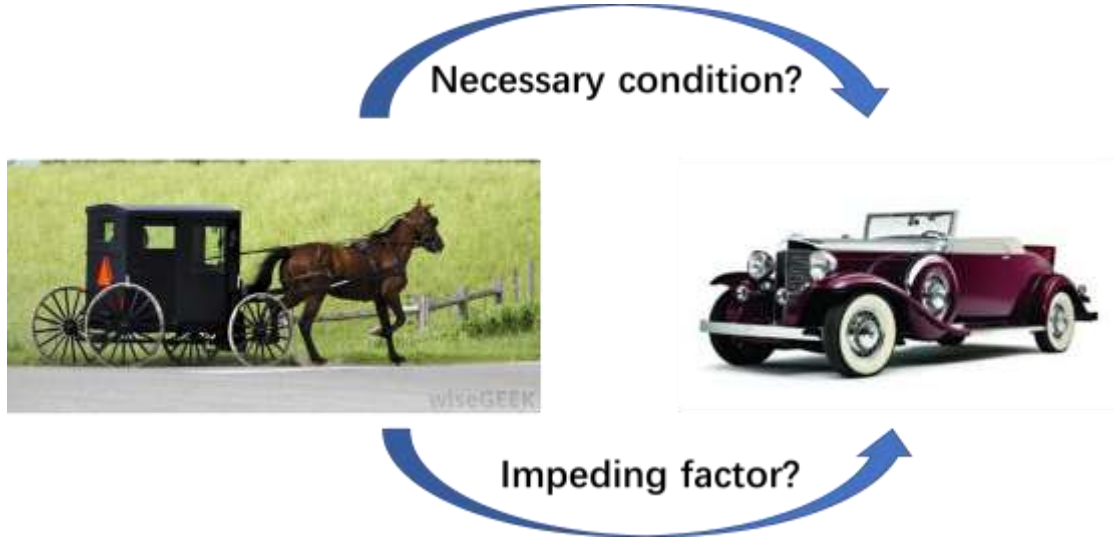
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- F3: some historian look at intellectual factors alone, while some at socio-economic factors alone

Fallacy 2: Example (Para. 40)

- The *Book of Changes* (易经) as an inhibiting factor?
- If Chinese scientists "were fully satisfied with an explanation they could find from the system of the *Book of Changes* they would go no further to look for mathematical formulations and experimental verifications of their scientific studies. Looking at the system of the *Book of Changes* in this light, one may regard it as one of the inhibiting factors in the development of scientific ideas in China."

Fallacy 2: no modern science because Chinese were satisfied with *Book of Change*

Horse and Buggy Metaphor (隐喻)



If we don’t have horse and buggy (马车) in China

- Fallacy 1: its absence in China made the invention of some analogue of the automobile impossible (because horse and buggy is a necessary condition for the invention of automobile)

If we find some analogue of horse and buggy in China

- Fallacy 2: horse and buggy is the inhibiting factor to invent automobile (because Chinese are satisfied with horse and buggy)

Summary

'Why the Scientific Revolution Did Not Take Place in China –
or Didn't It?'

This essay makes the case for two conclusions. First, why the scientific revolution did not take place in China is not a question that historical research can answer. **It becomes a useful question primarily when one locates the fallacies that lead people to ask it.** Second, a scientific revolution, by the criteria that historians of science use, did take place in China in the eighteenth century. It did not, however, have the social consequences that we assume a scientific revolution will have. (Para 1)

Scientific Revolution in China

China did undergo a scientific revolution because of the new science transmitted by Jesuit (耶稣会信徒) missionaries (传教士), like Matteo Ricci, in the late 16th and 17th centuries, but that it did not have the same ramifications as the Scientific Revolution did in Europe.

- Ming dynasty encounters with western science encouraged a revival of traditional Chinese astronomy and the rediscovery of ancient mathematics.
- The eclipse (日食) prediction contests in the late Ming court.



Matteo Ricci (利玛窦) and Xu Guangqi,
A. Kircher, *China Illustrata*. Amsterdam, 1667

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Critical thinking

- Critical thinking is skeptical without being cynical, evaluative without being judgmental, and purposefully focused on following reasons and evidence wherever they may lead.

- 批判性思维是怀疑而不是愤世嫉俗，是评估而不是评判，同时要有意识地专注于原因和证据，无论它们将引向何处。

Announcements

1. Finish reading Text 10.
2. Quiz next tutorial! (text 9 & 10)
3. TP and Online Forum II topics will be uploaded tonight.